

Project Proposal

Deep Learning Practical
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Proposed Idea

We propose to reproduce the paper **GPTSwarm: Language Agents as Optimizable Graphs** by Zhuge et al. (2024), published at ICML 2024 [1]. The paper introduces a framework that models LLM-based agents as directed acyclic computational graphs, where nodes represent operations (LLM inference, tool use, function calls) and edges define information flow between them. A key contribution is the automatic optimization of these graphs via two mechanisms: (1) edge optimization, which learns the inter-agent communication topology through a continuous relaxation and policy gradient updates, and (2) node optimization, which adapts individual node prompts based on task feedback. The authors validate their approach across four benchmarks: MMLU, Mini Crosswords, HumanEval, and GAIA.

An open-source implementation is available on GitHub. We will therefore maintain a full Reproduction Log in our final report, documenting hidden hyperparameters found in the codebase (e.g., edge sampling temperature, number of optimization steps per task), environment and dependency issues encountered, and any discrepancies between the paper’s claims and the actual implementation.

Experiment

We will reproduce the MMLU collaborative swarm optimization experiment (Section 3.1 of the paper), which is the primary benchmark. In this experiment, a swarm of GPT-based agents with different roles (e.g., domain experts, aggregator) is assembled as a composite graph, and the edge connectivity between agents is automatically optimized to maximize accuracy on MMLU multiple-choice questions. We will replicate both the fixed-topology baselines (fully connected, one-to-one) and the optimized swarm results, verifying the reported accuracy gains from graph optimization. This experiment directly demonstrates the core contribution of the paper and uses a standard, publicly available dataset.

Proposed Modification

We propose to replace the paper’s policy gradient-based edge optimization (REINFORCE) with an evolutionary algorithm (EA) for discovering optimal graph topologies [2]. Instead of maintaining continuous edge probabilities updated via gradient estimates, we evolve a population of discrete graph configurations. Each individual is a binary adjacency matrix defining a valid DAG; fitness is measured by task accuracy on a held-out validation split. We apply standard EA operators, tournament selection, single-point crossover on the flattened edge vector, and bit-flip mutation, while enforcing the DAG constraint after each variation step.

This modification is non-trivial for several reasons. First, it replaces the core optimization algorithm of the paper with a fundamentally different paradigm (gradient-free vs. gradient-based). Second, EAs explore the discrete topology space directly rather than through a continuous relaxation, potentially discovering qualitatively different communication structures. Third, it introduces a different exploration-exploitation trade-off. Finally, it enables a systematic ablation study, comparing population size, mutation rate, and generation budget against the original method, in terms of both final accuracy *and* number of LLM queries consumed, a cost dimension absent from the original paper.

A summary comparison of the two approaches is given in Table 1.

Table 1: Comparison of original edge optimization vs. proposed EA modification.

Aspect	GPTSwarm	Our modification (EA)
Search space	Continuous edge probabilities	Discrete binary graphs
Update rule	Gradient ascent (REINFORCE)	Selection + crossover + mutation
Exploration	Local, gradient-guided	Global, population-wide
Requires gradients	Yes	No

References

- [1] Zhuge, M., Wang, W., Kirsch, L., Faccio, F., Khizbullin, D., & Schmidhuber, J. (2024). GPTSwarm: Language Agents as Optimizable Graphs. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*. arXiv:2402.16823.
- [2] Eiben, A. E., & Smith, J. E. (2015). *Introduction to Evolutionary Computing* (2nd ed.). Springer.